Weekly Project Report – 3

Ahmedabad University
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Automated Multi-Parametric MRI Segmentation of Post-Treatment Glioma Sub-Regions Using Classical Machine Learning and Contour-Based Methods

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***Abstract*—In this report, the progress of Week 3 on automated brain tumor segmentation of multi-parametric MRI volumes using classical machine learning is summarized. This week, we accomplished data preprocessing, i.e. train/validation splitting, Z-score normalization, skull-stripped brain cropping, and started working on two different segmentation pipelines. Pipeline 1 is a combination of multi-scale Random Forest classifier and Chan-Vese Active Contour regions refinement. Multi-scale patch extraction, the Gray-Level Co-occurrence Matrix (GLCM) texture features, spatial context features, a Random Forest probability map, and Conditional Random Field (CRF) smoothing are also used to finalize the segmentation of Pipeline 2. The report outlines the preprocessing steps, pipeline architecture, details of the implementation and the week 4 plan.**

I. Introduction

Multi-modal MRI segmentation of brain tumors is an extremely important task which is applied in neuro-oncology because it can provide accurate volumetric measurements of tumor size and demarcation of sub-regions. Our project is based on the BraTS dataset [1] that has four MRI modalities T1, T1ce, T2, and FLAIR and expert-manual segmentation masks of the ground truth.

The current week is also the turning point of the literature review and system design (Week 2) and actual implementation. There are two parallel classical machine learning pipelines designed and currently being developed:

- **Pipeline 1:** Multi-scale Random Forest + Active Contour (Chan-Vese) refinement.

The two pipelines also work in a memory-efficient, patient-by-patient processing operation model to be able to execute in local machine execution without large intermediate datasets.

II. Dataset and Folder Structure

A. Dataset

Our dataset is the BraTS 2020 dataset [1] that consists of pre-operative multi-parametric MRI scans of glioma patients. Every patient instance presents four co-registered volumes of 3D:

- **T1:** Native T1-weighted scan.
- **T1ce (T1c):** Gadolinium contrast enhanced T1-weighted.
- **T2:** T2-weighted scan.
- **FLAIR:** Fluid Attenuated Inversion Recovery.

Ground truth segmentation labels are composed of three tumor sub-regions:

- Label 1: Necrotic and Non-enhancing Tumor Core (NCR/NET)
- Label 2: Peritumoral Edema (ED)
- Label 4: Enhancing Tumor (ET)

B. Folder Structure

Following preprocessing the working directory becomes arranged in the following way:

```
Brats_Project/
 additional_training/
 training/
 validation_training/
 cropped_dataset/
 cropped_train/
    <patient_id>/
        t1.npy
        t1c.npy
        t2.npy
        flair.npy
        seg.npy
 cropped_val/
    <patient_id>/
        t1.npy
        ...
        seg.npy
 BraTS_Working_Pipeline.ipynb
```

A 3D MRI volume is saved in each `.npy` file. The preprocessing pipeline generates the `cropped_train/` and `cropped_val/` directory without generating any redundant medium datasets.

III. Week 3 Progress: Data Preprocessing and Model Development

During Week 3, the focus of the project shifted from system design and literature review toward the implementation of the segmentation pipeline. Several key stages were completed including dataset preprocessing, voxel-level feature extraction, Random Forest model training, and spatial refinement using Markov Random Fields (MRF).

A. Dataset Cropping and Preprocessing

The original BraTS dataset contains full 3D MRI volumes with large background regions outside the brain. Processing the full volumes would significantly increase computational cost without contributing useful information.

To address this, the dataset was first converted into cropped brain volumes by computing a tight bounding box around the non-zero brain voxels across all four MRI modalities.

This preprocessing step provided several advantages:

- Removal of large empty regions surrounding the skull
- Significant reduction in computational cost
- Improved efficiency for feature extraction and model training

- Reduced memory requirements for processing 3D volumes

The cropped dataset was then stored as NumPy arrays for faster loading during training.

B. Dataset Split

After cropping the dataset, the patient folders were divided into two subsets:

- **Training Dataset:** stored in `cropped_train/`
- **Validation Dataset:** stored in `cropped_val/`

Each patient directory contains the following files:

```
<patient_id>/
  t1.npy
  t1c.npy
  t2.npy
  flair.npy
  seg.npy
```

These NumPy files store the normalized 3D MRI volumes and segmentation masks.

C. Z-Score Intensity Normalization

MRI intensity distributions vary significantly between patients and scanners. To standardize intensity values, Z-score normalization was applied independently to each modality.

For a given MRI volume V with brain voxels Ω:

$$V_{norm}(x) = \frac{V(x) - \mu_\Omega}{\sigma_\Omega}, \quad x \in \Omega \tag{1}$$

where μ_Ω and σ_Ω represent the mean and standard deviation computed only over non-zero brain voxels. Background voxels remain unchanged.

This normalization ensures that voxel intensities across different patients become comparable.

D. Multi-Resolution Feature Extraction

Multi-resolution contextual neighborhoods were used to extract voxel-level features following normalization.

Features from all four MRI modalities (T1, T1ce, T2, and FLAIR) were calculated for every voxel. There were two contextual windows utilized:

- $3 \times 3 \times 3$ neighborhood capturing fine local texture
- $5 \times 5 \times 5$ neighborhood capturing broader regional context

The following features were computed for each modality:

- Raw voxel intensity
- Abnormality score relative to brain tissue statistics
- Gradient magnitude
- Local mean
- Local variance

These characteristics enable the classifier to identify local structural patterns in brain tissue as well as variations in intensity.

E. Voxel Sampling for Training Dataset

Millions of voxels make up each MRI volume. Direct training on every voxel would significantly extend training time and surpass memory constraints.

A balanced voxel sampling strategy was used to address this issue.

For every patient:

- Tumor voxels were identified using the segmentation mask.
- Normal brain voxels were identified from non-tumor regions.
- Approximately 3000 tumor voxels and 3000 normal voxels were randomly sampled.

The Random Forest model is kept from becoming biased in favor of the majority non-tumor class thanks to this balanced sampling.

The final training dataset was created by combining the sampled voxels from every patient:

- `X_train_3x3`: features from $3 \times 3 \times 3$ neighborhoods
- `X_train_5x5`: features from $5 \times 5 \times 5$ neighborhoods
- `y_train`: corresponding tumor labels

To maintain memory efficiency, each patient was processed sequentially and large intermediate arrays were immediately released.

F. Random Forest Model Training

Using the sampled voxel dataset, Random Forest classifiers were trained to distinguish tumor and non-tumor voxels.

Two contextual models were designed:

- **3×3 Local Context Model**
- **5×5 Regional Context Model**

During Week 3, the 3×3 model was fully trained using the following parameters:

- Number of trees: 100
- Maximum depth: 15
- Random seed: 42

The trained model was saved using the `joblib` library to allow efficient reuse during inference.

Due to computational limitations, only the 3×3 model has been trained and evaluated so far. The 5×5 model will be trained and integrated during Week 4.

G. MRF-Based Segmentation Refinement

The Random Forest model creates a voxel-wise tumor probability map. However, these predictions may include isolated noisy voxels.

To improve spatial consistency, we added a Markov Random Field (MRF) refinement step.

MRF smoothing encourages neighboring voxels to have similar labels. This creates more coherent tumor

regions. The refinement process updates voxel labels in cycles using both the classifier probability and the labels of neighboring voxels.

After the MRF refinement, we used morphological operations like binary opening and hole filling to eliminate small artifacts and create smoother tumor boundaries.

H. Segmentation Visualization

To evaluate the segmentation pipeline, predictions were generated for unseen validation patients[cite: 117]. Figure 1 shows an example result: the left panel shows the original FLAIR MRI slice, the middle panel shows the expert ground truth tumor mask, and the right panel shows the predicted tumor segmentation after Random Forest classification and MRF refinement[cite: 118].

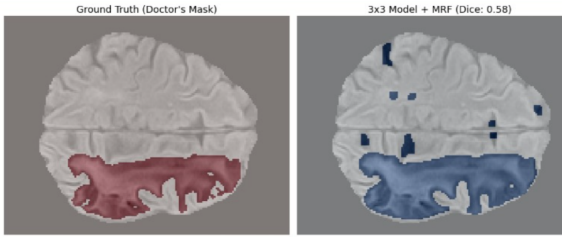

Fig. 1: Example segmentation result showing the original MRI slice, the ground truth tumor mask, and the predicted tumor segmentation produced by the Random Forest model with MRF refinement[cite: 137].

The estimated segmentation is effective in capturing the large tumor region and smoothness of spatial aspect following refinement[cite: 138]. Some of the final measures of the displayed patient (BRATS-GLI-02069-101) are Dice Score of 0.5839, Sensitivity of 0.9945, and Specificity of 0.9691[cite: 119].

IV. Plan for Week 4

In Week 4, the project will focus on improving the segmentation pipeline and conducting a more rigorous evaluation.

The following tasks are planned:

- **Dataset Split Improvement**
  The dataset will be reorganized into a standard machine learning split consisting of:
  - 75% training data
  - 15% validation data
  - 10% test data

  This will allow unbiased evaluation of the segmentation models.
- **Training of the 5×5 Context Model**
  The 3×3 model will be already trained, and the 5×5 model will be trained in Week 4. This model represents bigger spatial patterns and is likely to enhance the performance of segmentation.
- **Multi-Scale Feature Fusion**
  The outputs of the 3×3 and 5×5 models will be combined to produce a multi-scale tumor probability map that leverages both local and regional contextual information.
- **Hyperparameter Optimization**
  Further experiments will be conducted to optimize:
  - number of Random Forest trees
  - maximum tree depth
  - MRF smoothing parameters
- **Full Validation Evaluation**
  The segmentation pipeline will be evaluated across the entire validation dataset using:
  - Dice Similarity Coefficient
  - Sensitivity
  - Specificity

V. Conclusion

The week was devoted to the commitment of the entire preprocessing pipeline and the beginning of the first segmentation method of automated sub-regions segmentation of glioma. The preprocessing step (Z-score normalization and bounding-box cropping) makes sure that all patient scans have equal quality of input and enhances the accuracy of the following tasks in its segmentation.The pipeline 1 relies on the robustness and simplicity of active contour-based refinement of boundaries. In this method, the outline around tumor masses is developed through repeated procedures of changing the boundaries using image gradients and local data. Through this type of refining the tumor edges, the process is expected to deliver a more accurate depiction of the sub-regions of glioma in MRI images.In sum, the presented pipeline focuses on a classic boundary-evolution segmentation approach, which offers the structured and computationally efficient way of localizing tumor boundaries in medical imaging.

References

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